$f(x)$ is well-defined $(x_0-\theta, x_0)\cup(x_0, x_0+\theta)$ $\forall \varepsilon>0$, $\exists \delta>0$,
such that $|x-x_0|<\delta \Rightarrow |f(x)-A|<\varepsilon$.
we say $\lim_{x\to x_0} f(x) = A$

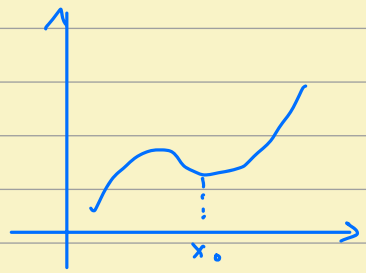
$$f(x)=\frac{\sin x}{x} \qquad \lim_{x\to 0} f(x) = 1$$

$$\lim_{x\to+\infty} \sin x \qquad \lim_{x\to\infty} x^2 = +\infty \qquad \sin x \sim x \quad x\to 0$$

$$x \sim x+1 \quad x\to\infty \quad \lim_{x\to\infty}\frac{x}{x+1} = 1$$

$$\infty+\infty \qquad \infty-\infty \qquad \frac{\infty}{\infty}$$

Continuous functions

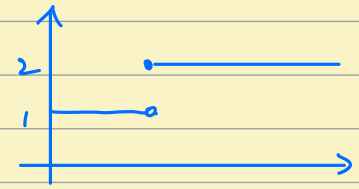


Defn (Continuous function) Suppose that $f(x)$ is well-defined on $(x_0-\delta, x_0+\delta)$
If

$$\lim_{x\to x_0} f(x) = f(x_0)$$

then we say that f is continuous on x_0.

① $f(x_0)$ is defined
② $\lim_{x\to x_0} f(x)$ exists.
③ $\lim_{x\to x_0} f(x) = f(x_0)$

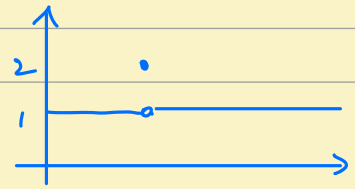


$$f(x)=\begin{cases} 2 & x\geq 2 \\ 1 & x<1 \end{cases}$$

Defn (ε-δ-definition) For any $\varepsilon>0$, there exists $\delta>0$, such that $|x-x_0|<\delta \Rightarrow$

$$|f(x)-f(x_0)|<\varepsilon.$$

$\hookrightarrow \lim_{x\to x_0} f(x) = f(x_0)$
$f(x_0)$ is defined.



$$f(x)=\begin{cases} 2 & x=2 \\ 1 & \text{otherwise} \end{cases}$$

E.g. $D(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$

$|x \cdot D(x)| \le |x| \cdot \underbrace{|D(x)|}_{\le 1}$
$\le |x|$

Function $f(x) = x \cdot D(x)$ is continuous on $x = 0$.

$f(0) = 0 \cdot 1 = 0$ for any $\varepsilon > 0$, take $\delta = \varepsilon$

$$|f(x) - f(0)| = |x \cdot D(x) - 0| \le |x - 0| < \varepsilon$$

Defn (left, (right) continuous)

If a function f is defined on $(x_0 - \delta, x_0]$ $[x_0, x_0 + \delta)$, we say f is left-continuous if

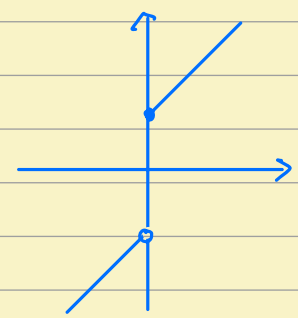
$$\lim_{x \to x_0^-} f(x) = f(x_0)$$

right-continuous

$$\lim_{x \to x_0^+} f(x) = f(x_0)$$

Thm A function f is continuous on x_0 if and only if f is both left continuous and right continuous at x_0.

E.g. $f(x) = \begin{cases} x+2 & x \ge 0 \\ x-2 & x < 0 \end{cases}$



$$\lim_{x \to 0^+} f(x) = 2 = f(0)$$

$$\lim_{x \to 0} f(x) = -2 \ne f(0)$$

Discontinuous function

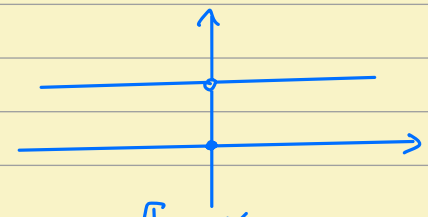
If function f satisfies one of the following:

① f is not defined at x_0

② f has no limit at x_0

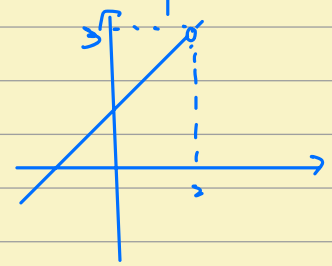
③ f is defined at x_0 and $\lim_{x \to x_0} f(x)$ exists, but $\lim_{x \to x_0} f(x) \ne f(x_0)$

Removable discontinuity 可去间断点



E.g. $f(x) = |\operatorname{sgn} x|$, $x = 0$

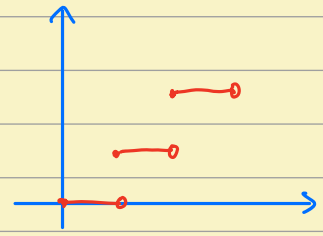
$$g(x) = \begin{cases} |\operatorname{sgn} x|, & x \neq 0 \\ 1, & x = 0 \end{cases}$$



$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

$$g(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & x \neq 2 \\ 3 & x = 2 \end{cases}$$

Jump discontinuity 跳跃间断点



E.g. $f(x) = [x]$

Both left and right limits exist but not equal.

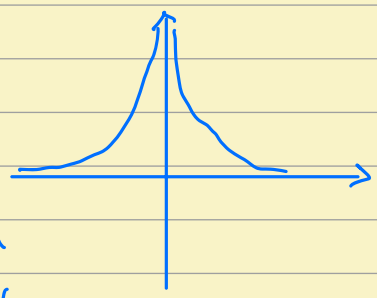
Type I discontinuity: Removable discontinuity & Jump discontinuity

Both left and right limits exist.

Type II discontinuity: At least one-side limit does not exist

E.g. $f(x) = \frac{1}{x}$

$f(x) = \frac{1}{x^2}$



$$D(x) = \begin{cases} 1 & x \text{ is rational} \\ 0 & x \text{ is irrational} \end{cases}$$

Continuous function on Interval

Function f is said to be continuous on (a,b), if f is continuous ant every point on (a,b)

Function f is said to be continuous on $[a,b]$, if f is continuous on (a,b), and f is right/left continuous on a/b.

$$\lim_{x\to b^-} f(x) = f(b) \qquad \lim_{x\to a^+} f(x) = f(a)$$

Properties of continuous function

Thm (Boundedness) Let f be continuous on $[a,b]$, then f is bounded.

E.g. $f(x) = \frac{1}{x}$ is not bounded on $(0,1)$

Thm (Boundedness) Let f be continuous at x_0, then there exists $\delta > 0$ such that f is bounded on $(x_0-\delta, x_0+\delta)$

Thm. Let f be continuous at x_0, if $f(x_0) > 0$, then there exists $\delta > 0$, such that $f(x) > 0$, for any $x \in (x_0-\delta, x_0+\delta)$.

Thm. Let f, g be continuous at x_0, then $f \pm g$, $f \cdot g$. f/g, $g(x_0) \neq 0$ are continuous at x_0

Thm. Let f be continuous at x_0, g be continuous u_0, $u_0 = f(x_0)$.
$g(f(x))$ is continuous at x_0.

Proof: Since g is continuous at u_0,

$\forall \varepsilon > 0, \exists \delta_1 > 0, \quad |u - u_0| < \delta_1 \Rightarrow |g(u) - g(u_0)| < \varepsilon.$

Since f is continuous at x_0,

$\forall \delta_1 > 0 \ \exists \delta > 0, \quad |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \delta_1$

$\Rightarrow \forall \varepsilon > 0, \exists \delta > 0, \quad |x - x_0| < \delta \Rightarrow |g(f(x)) - g(f(x_0))| < \varepsilon.$

$$\lim_{x \to x_0} g(f(x)) = g\left(\lim_{x \to x_0} f(x)\right) = g(f(x_0))$$

E.g. $\lim_{x \to 1} \sin(1 - x^2) = \sin\left(\lim_{x \to 0} (1 - x^2)\right) = \sin 0 = 0$

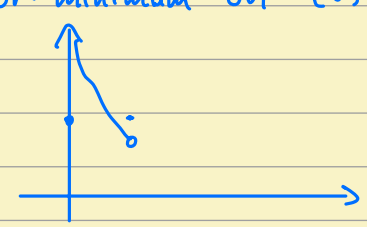
Defn (Maximum, minimum) Let f is well-defined on I, if there exists $x_0 \in I$, such that for any $x \in I$, we have

$$f(x_0) \geq f(x). \quad f(x_0) \leq f(x)$$

then we say f has maximum (minimum) on I, and the maxima (minima) of f on I is $f(x_0)$

E.g. $f(x) = x$ has no maximum or minimum on $(0,1)$

$$f(x) = \begin{cases} \frac{1}{x} & x \in (0,1) \\ 2 & x = 0 \text{ or } x = 1 \end{cases}$$



Thm. (Minimum-maximum theorem) If f is continuous on $[a,b]$, then f has maxima and minima on $[a,b]$.

Lemma. If f is continuous on $[a,b]$, then f is bounded on $[a,b]$.

Proof: Suppose by contradiction $f(x)$ is not bounded on $[a,b]$, there exists $x_n \in [a,b]$, $f(x_n) > n$, $\lim_{n\to\infty} f(x_n) = +\infty$

Since $\{x_n\} \subset [a,b]$ by the compactness we have that there is a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that $\lim_{k\to\infty} x_{n_k} = x_0$

$a \le x_{n_k} \le b \Rightarrow a \le x_0 \le b$ then we have $f(x)$ is continuous on x_0

$\lim_{n\to\infty} f(x_n) = +\infty$, then contradiction. Hence $f(x)$ is bounded.

Proof (Minimum-maximum theorem) Write

$$\sup_{x\in[a,b]} f(x) = M$$

To prove $\xi \in [a,b]$, such that $f(\xi) = M$. Suppose by contradiction that $f(x) < M$ for all $x \in [a,b]$. Let

$$g(x) = \frac{1}{M - f(x)} \qquad x \in [a,b]$$

then, by the above lemma, we have $g(x)$ is bounded above.

$$0 < g(x) = \frac{1}{M-f(x)} \le G$$

$$\Rightarrow \frac{1}{M-f(x)} \le G \Rightarrow f(x) \le M - \frac{1}{G}$$

Since M is the supremum of f on $[a,b]$. Contradiction. Then there exist $\xi \in [a,b]$, such that $f(\xi) = M$.